

JIWON YANG 양지원

DESIGN UNDERGRADUATE | AFFECTIVE & MULTISENSORY INTERACTION | HCI

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RESEARCH INTERESTS

Designer-researcher working at the intersection of **affective computing**, **multisensory interaction**, and **HCI**. My work translates non-verbal sensory experience (particularly sound and touch) into personalized cross-modal representations, using both interactive machine learning and physical computing. I am drawn to designing natural, human interfaces that bridge subjective perception and computational systems, with a focus on accessibility and creative expression.

EDUCATION

Seoul National University, B.F.A. in Design 2023 to Feb 2028 (expected)

- Major GPA: 4.0 / 4.3
- **HCI & Interaction:** Human Behavior and Design (A+), Object Interaction Design (A+), Product Service Design (A+), UI Design Programming (A0), Media Design Programming (A0)
- **AI & Engineering:** Design for Machine Learning (A+), Extended Reality Design (A+), Data Visualization (S), Introduction to AI (S)

University of Sydney, Exchange Student, Interaction Design 2024 (Sem. 2)

RESEARCH EXPERIENCE

ASC Lab, KAIST, Student Intern Sep 2026 to Present

Independent Research, Researcher (First Author) Feb 2026 to Jul 2026

"Beyond Parentheses: Personalizing Graphical Sound Captions through Interactive Machine Learning"

- Led the entire project end-to-end: problem framing, system design, implementation, user study, and manuscript writing.
- Built the **IML Audio Workstation**, a dual-head (multi-task) neural network mapping psychoacoustic features to personalized 2D motion graphics, rendered in WebGL / Three.js.
- Designed a **"Listen / Sculpt / Train"** human-in-the-loop workflow for accessible, personalized sound captioning (incl. Deaf and hard-of-hearing viewers); pilot study (n=4) raised perceptual agreement from 3.2 to 6.4 on a 7-point scale.

Medical AI Lab (IMSI), Seoul National University, Undergraduate Research Intern Feb 2026 to Aug 2026

- Implemented and experimented with medical-imaging models and tasks in PyTorch.
- Led data visualization and paper-figure design for medical imaging research, bridging design and ML.

TEACHING

CIT Code Academy, AI Design Instructor 2023 to Present

- Teaching AI design to 10+ students in grades 10 to 12, sustained across three years.
- The academy is a partner in ITU's **AI Skills Coalition** (AI for Good), the UN initiative for open AI education launched in January 2025.

SELECTED PROJECTS

Playground: A Line Between Us 2026

- Interactive installation that reads a visitor's touch gesture through a sparse sensor grid and "answers" by drawing back in sand, translating the feel of a gesture (pressure, pace) rather than copying it. Tools: physical computing (sensor grid), actuated drawing machine. **Selected for SIGGRAPH Asia 2026 Art Gallery.**

Chorus, Team 2025

- Medieval-organ-inspired interactive artwork: light sensors detect a visitor's playing gestures to control four "door" structures, modulating emitted light and four-track audio levels so the visitor conducts an orchestral harmony in real time.
- Role: overall system design and code implementation (teammate: modeling, fabrication & installation). Tools: light sensors, sound/lighting control.

SendLove, Individual 2025

- Interactive sculpture: layered acrylic panels form the waveform of a recorded voice saying "사랑해" (I love you); an ultrasonic sensor detects a "sending" hand gesture, lighting per-panel LED strips in sequence to evoke a message being sent. Tools: ultrasonic sensor, LED strips, sound visualization.

PREPRINTS & MANUSCRIPTS

- Anna Jung, Kyeonghun Kim, Youngung Han, Eunseob Choi, **Jiwon Yang**, Ken Ying-Kai Liao, Hyuk-Jae Lee, Nam-Joon Kim. "ProsMAE: Multi-Source MAE Pretraining for ISUP Grade Classification." **Accepted to IEEE APCCAS 2026 (Poster Session).**

Contribution: Data visualization and figure design.

- Kyeonghun Kim, Jaehyung Park, Youngung Han, Anna Jung, Seongbin Park, Sumin Lee, **Jiwon Yang**, et al. "MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy." arXiv preprint arXiv:2604.00537, 2026. arxiv.org/abs/2604.00537

Contribution: Research design support, manuscript writing, and figure design.

- Insung Hwang, Kyeonghun Kim, Youngung Han, et al., **Jiwon Yang**, et al. "LeafFireNet: Hyperspectral Leaf Dryness Classification for Wildfire Risk Prediction Using Temporal SWIR Spectral Signatures." Submitted to IEIE Summer Conference 2026 (under review).

Contribution: Research assistance.

EXHIBITIONS (JURIED)

- **Playground: A Line Between Us.** Selected for SIGGRAPH Asia 2026 Art Gallery. Advised by Chang Hee Lee. Role: concept, interaction design, system implementation, and fabrication.

SKILLS

DESIGN (2D)	Photoshop, Illustrator, After Effects, Figma
DESIGN (3D)	Rhino, Blender, Fusion 360
ML / AI	PyTorch (model training), Generative AI (Diffusion Models)
COMPUTER VISION	OpenCV
PHYSICAL COMPUTING	Arduino, sensors, actuator control
XR / GRAPHICS	Unity-based XR, WebGL / Three.js
WEB	HTML, CSS, JavaScript

LEADERSHIP & ACTIVITIES

- XREAL (XR Society), SNU. Design Lead, 2025 Summer to 2026 Summer
- SNU Design Alliance. Member, 2025
- ISEA 2025. Student Volunteer, 2025
- Gwangju Design Biennale Challenge, 2025

CERTIFICATIONS & LANGUAGES

- NVIDIA DLI. Generative AI with Diffusion Models, 2025
- IELTS. Overall 7.5, 2024